

Uncovering Non-Linear Joint Tail Events in Sector Pairs Using Mutual Information



Abstract

Growing interconnectivity across different sectors of interest to the financial markets has led to an increasing number of hidden relationships across macro and microeconomic factors. With rising data complexity, traditional methods within risk analysis and portfolio management that rely on finding linear relationships are not as effective at finding critical non-linear patterns between variables. However, the advancements of non-linear methods mean that we are able to capture relationships between variables that were hidden beforehand. This research paper looks to investigate whether mutual information - a non-linear dependency measure - can find hidden relationships during joint tail events (at the 5% quantile) in sector pairs. We then select the three pairs with a large absolute difference their respective mutual information measure and Pearson correlation and analyse their spread behaviour during a joint tail event using a Kernel Density Estimation (KDE). We find that mutual information is more effective than correlation at revealing asymmetric spread behaviour, and paired sector crashes under extreme market conditions.

1 Introduction

In the increasingly interconnected financial landscape and growing complexity of financial products, the use of traditional methods have been outgrown. Traditional methods that had emphasised linear relationships had grown to become outdated, as they were realised to be unable to capture the hidden non-linear relationships amongst variables. Mutual information acts as a measure that is able to uncover these relationships without relying on the same assumptions that correlation relies on, such as normality. Furthermore, correlations insufficient ability to detect dependencies and co-movement in times of market stress, otherwise known as tail events is motivation for a step towards mutual information. This project seeks to explore the extent to which mutual information is able to uncover joint tail dependencies for sector pairs and their co-movement, which would be hidden to correlation measurements.

The project is structured in the following way. The first section details a literature review, exploring current research on the strengths of mutual infor-

mation and its use in joint tail analysis and sector pair trading. Second is the methodology. This section details the pre-processing and cleaning of the data, measuring correlation and mutual independence, identifying pairs, and the kernel density estimation to build the final distribution for each notable pair. The results section analyses each pair, and details the hidden, non-linear relationships we find between sectors that Pearson correlation cannot find. The next section on discussion looks on possible directions this research can expand towards, and then finalising on the conclusion.

1.1 Literature Review

Correlation is a linear dependence measure that quantifies the extent to which two variables move in relation to each other. Pearson’s correlation coefficient is similarly defined as the normalised covariance. In the use of correlation in a financial time series, there is a strong assumption that the data will usually exhibit normally distributed returns, however this is not realistic. As Embrechts et al. (2002) notes, the ”inclusion of non-linear derivative products invalidates many of the distributional assumptions underlying the use of correlation”, highlighting the limitations of linear correlation in modern financial analysis. Furthermore, (Embrechts et al., 2002) demonstrates how two different distributions with the same correlation coefficient can both exhibit different behaviours in the tails. Effectively displaying the underlying problems of correlation in modelling tail events.

In an attempt to circumvent the limitations of linear correlation, much analysis has shifted towards information theory, and its derived models. Shannon (1948) introduced the concept of entropy, originating from information theory as a measure of uncertainty. In financial time series context, a high entropy means you are less likely to predict the next variable. This is key in the utilisation of entropy-derived methods, as these methods are able to detect hidden relationships that were difficult to uncover beforehand (Darbellay and Wuertz, 2000; Dionisio et al., 2004). For example, Maasoumi and Racine (2002) found that the ”metric entropy is capable of detecting non-linear ’affinity’” between two different time series. He demonstrated this noting the inadequacy of correlation in the cases of a ”non-Gaussian, linear, and nonsymmetric circumstances found in financial processes”. Thus, there is greater emphasis on the data structure itself, rather than its underlying distribution.

Mutual information is another concept that was born out of entropy. It is an information dependency measure that assigns a quantity to the amount of information shared between two variables. Unlike Pearson correlation, which captures only linear relationships, mutual information accounts for both linear and non-linear dependencies. In finance, mutual information has been used to detect hidden relationships between asset classes, and improve portfolio diversification strategies. Previous studies have shown that mutual information can detect comovement patterns that are missed by traditional correlation—especially under volatile conditions, such as Guo et al. (2018).

In continuous data, such as a financial time series, the mutual information

formula is:

$$I(X; Y) = \int \int p(x, y) \log \left(\frac{p(x, y)}{p(x) p(y)} \right) dx dy$$

In terms of entropy, it would then become:

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

2 Methodology

2.1 Pre-Processing

Through an exploratory data analysis of all the data, we had found the data to likely represent a set of closing prices time series for the tickers, where each sector was split into separate files. There were 12 sectors ranging from financials to technology to consumer goods amongst many others. To find matching dates for our return series we aligned tickers with the yahoo finance returns. In all files we found matching dates for our time series, but there were extra rows in which we could not find a match. These extra rows were removed and the finalised set of data had dates ranging from 4th January 2006 to 9th May 2014. The final result was 1252 different stocks, with no missing values.

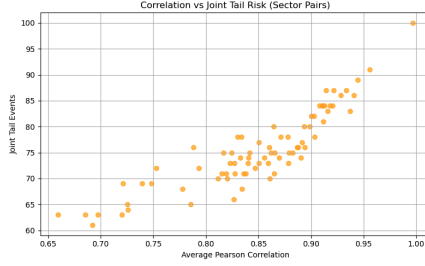
To prepare the data for use in this project, we use a dictionary in which the keys are the sector and the values are a dataframe that contain all the tickers and their respective closing prices. We then transform the returns into log returns rather than the closing prices as to optimise our dataset. Log returns enforce stationarity as compared to raw prices which can show trends across time. Having stationarity in the returns ensures we really measure the sectors co-movement rather than their trends that are shared. After finding the sector log returns, the average is taken across all tickers for each day to create a return time series for each sector to represent the average sector performance.

2.2 Measuring Dependency with Mutual Information and Correlation

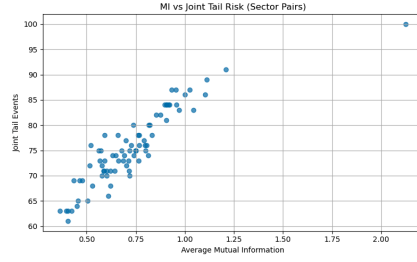
As previously mentioned, the research is meant to act as a comparison in the effectiveness of studying co-movement in sector pairs in joint tail events by two metrics: mutual information and Pearson's correlation coefficient. We loop through all sector pairs with a rolling window of 60 days to ensure the capture of changes in scores in times of crisis and non-crisis. This allows us to evaluate how the two sectors interact in differing market regimes. The loop finds the mutual information and correlation for all pairs in the window time frame and stores the result in a data frame. This approach allows us to understand how the dependency evolves over time in comparison to a solely linear dependency across all sector pairs.

The plot below shows the average mutual information and correlation for sector pairs in periods where their returns were in the bottom 5% quantile. While

both the correlation and mutual information measures show a general positive trend, the mutual information is much steeper, and tighter. This suggests that mutual information is able to capture more of the dependency structure than the Pearson correlation is able to, specifically non-linear events.



(a) Correlation Graph



(b) Mutual Information Graph

2.3 Pair Selection

The objective of the pair selection is to select sector pairs that show a high dependency gap - defined as the difference between mutual information and correlation, as a way to isolate for non-linear dependence. This measure gives the non-linear information shared between sectors, that could not previously be explained by the linear correlation.

We then define a tail event when a sector experiences daily closing returns below its 5th percentile threshold. We identified occurrences in which each sector's returns fell below this threshold, and counted the days where the sector pairs crashed simultaneously. Then taking the average mutual information and Pearson correlation for each sector pair and compiling them into a dataframe, which also included each sector pairs dependency gap, and the days spent together in a tail event.

The finalised dataframe was then ranked in descending order by dependency gap, and three pairs were then chosen for further analysis. These pairs are the basis for our Kernel Density Estimation analysis as to visualise asymmetric behaviours in joint tail events.

The plot below shows a dependency gap matrix:

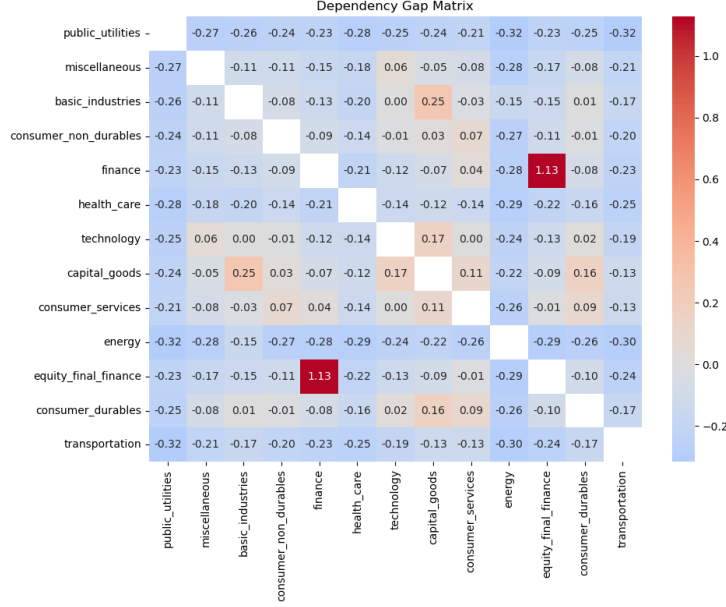


Figure 2: Dependency Gap Matrix

The sector pairs with a dependency gap value of 0.10 or higher are of interest to our analysis. This effectively means that there are dependencies being picked up by the mutual information measure, that were not noticed by the correlation. One notable outlier is the 'finance' and 'equity_final_finance' sector pair due to its inflated dependency gap value of 1.13. These two datasets feature a high overlap in tickers within them, in which they have near identical time series data for the respective tickers. Inclusion of this pair in the further analysis would make redundant the measure of a dependency score, therefore it is excluded.

2.4 Kernel Density Analysis

Kernel densities are a non parametric methodology to estimate a probability distribution function using a finite set of observations. The strengths of this method lie in that it holds no assumptions in the underlying distributions of the data, allowing for flexibility in modelling real data. This is key for financial time series analysis, as returns data usually will exhibit skewness, excess kurtosis, and tail risk, that are difficult to model to real-world standards.

The KDE is used to estimate a distribution for the returns of sector pairs in joint tail events where they occur in the same time period - these will be the observed points that build the distribution. Each observation acts as a weighting function centred at its point. These are then combined to produce a smooth and continuous estimate of the distribution. The formula for the KDE is below:

$$\hat{f}(x) = \frac{1}{qh} \sum_{k=1}^q K\left(\frac{x - x_k}{h}\right)$$

In this project, we had chosen the Gaussian kernel, which is the default for the 'SciPy' Python library, due to its ensured smoothness. A kernel function dictates the shape of each observed point and thus, its contribution to the overall density estimate. It is key to note - the shape of the kernel does not hold any holistic influence over the shape of the final probability distribution once we combine all observations.

3 Results

In this section we detail the findings of the three selected sector pairs with large dependency gaps. By plotting the KDE, alongside a line plot that shows the rolling average's of the mutual information and the correlation for the sector pair, we are able to identify key behaviours of each pair. Firstly, we show that the return spread distributions clearly exhibit asymmetric behaviours, in that one sector consistently underperforms the other during tail events. Secondly, there is variability in the spreads. Some exhibit a tighter distribution indicating greater stability during tail events whereas broader KDE's may indicate greater volatility in the co-movement of the sectors. Finally, we introduce a minimum spanning tree as to understand why the Capital Goods sector is frequently appearing as one of the sectors in pairs with a high dependency gap.

3.1 Basic Industries and Capital Goods

The KDE plot reveals a clear bi-modal structure, in which the relation between the two sectors changes. The peak in the left tail at a spread of -0.015 tells us that Basic Industries underperform Capital Goods in downturns. This decline is not as sharp, meaning a higher density so this scenario is not uncommon. However, in the right tail, there is a higher peak, a faster downturn, and a tighter spread. These all indicate the rare event in which Basic Industries outperform Capital Goods. In tail events, this is a rare occurrence. The asymmetry shown by this KDE is a clear representation of the non-linear dependencies that correlation is unable to capture.

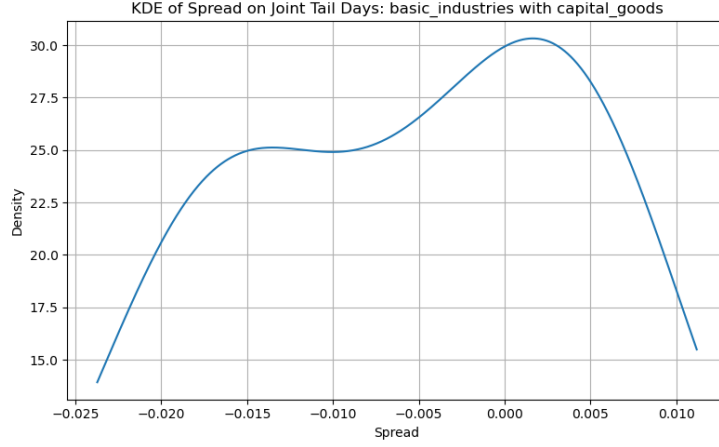


Figure 3: KDE: Basic Industries and Capital Goods

The two different peaks in this KDE may indicate differing behaviours under different market regimes where co-movement diverges. The left tail indicates that Capital Goods performs even worse during specific crashes so the sectors co-movement now diverges. The lack of non-uniform behaviour between the sectors could also indicate changing behaviours based on external macroeconomic factors, possibly changing based on the type of financial crash. Examples of this being COVID-19 and the Global Financial Crisis. Thus, the line graph below shows how across the entire time range, Pearson correlation does not pick up the changes in dependency that mutual information can.

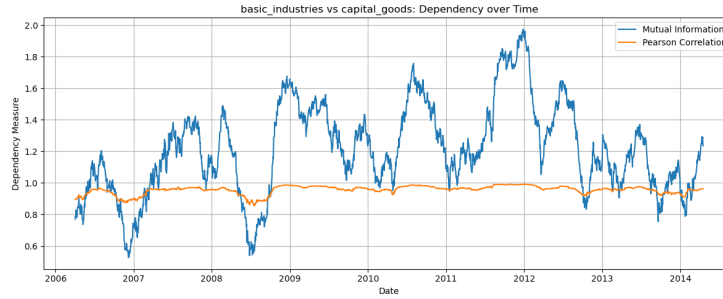


Figure 4: Line plot: Basic Industries and Capital Goods

3.2 Technology and Capital Goods

This sector pair produces a positively centred KDE with a spread centred around approximately 0.013 (near zero), indicating that Technology outperforms Capital Goods during tail events. There is strong and directional co-movement between this sector pair given the tighter spread across the KDE inferring sta-

bility in the sector pair behaviour during financial crashes. This means that the behaviours of the pairs are somewhat predictable and not volatile so are a strong candidate for being used in hedging during a tail event. Clearly, correlation cannot give indication as to the directionality of the two sectors due to the hidden relationship which is displayed in the line plot below.

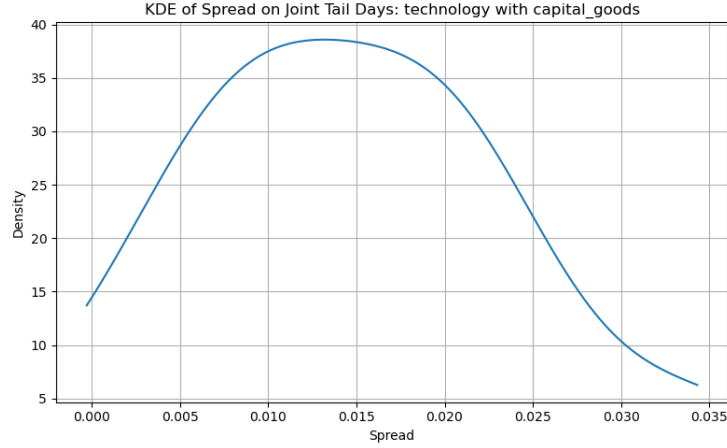


Figure 5: KDE: Technology and Capital Goods

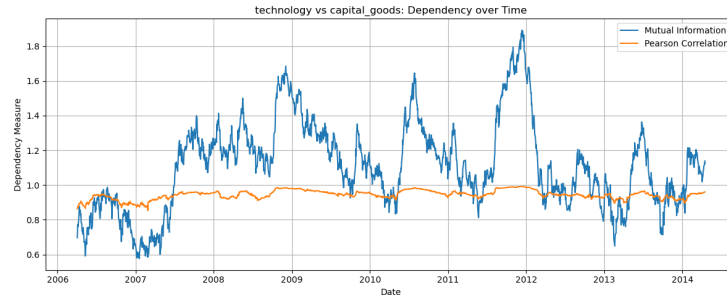


Figure 6: Line plot: Technology and Capital Goods

3.3 Capital Goods and Consumer Services

Our final KDE plot indicates a slight asymmetry in which Capital Goods are underperforming again relative to Consumer Services, however not as large an underperformance as the two previous sector pairs. There is a left skew, in which the left tail ranges from -0.02% to -0.007% , with peak density at a negative spread of around -0.005% . There is a large density in the peak area of the KDE indicating slight underperformance by Capital Goods, but the length of the left tail indicates a general consistency in which Consumer Services outperforms

Capital Goods in tail events. The right tail indicates that Capital Goods rarely outperforms Consumer Services and has a much lower weight in crash periods.

The line plot similarly shows the level of hidden information that is not captured by the correlation.

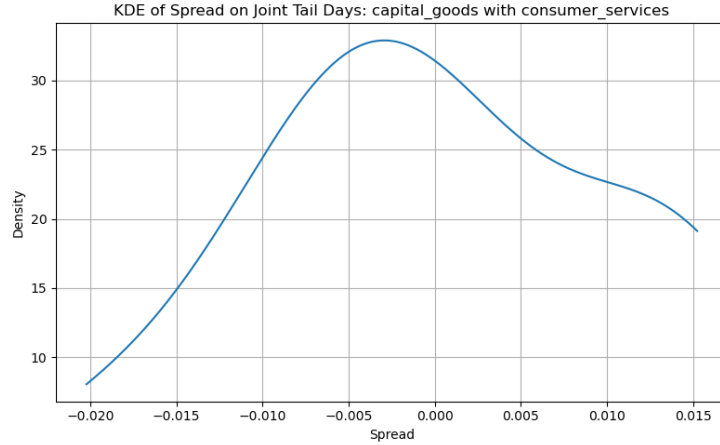


Figure 7: KDE: Capital Goods and Consumer Services

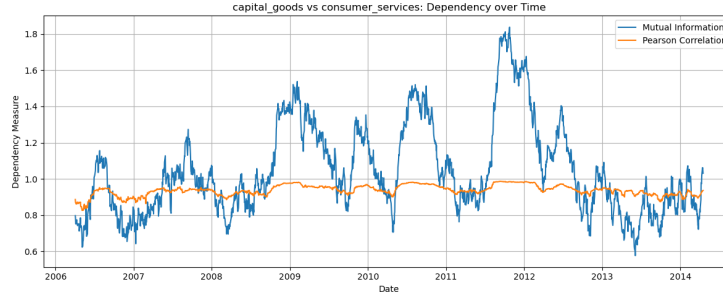


Figure 8: Line Plot: Capital Goods and Consumer Services

3.4 KDE Validation

To ensure our KDE's are valid approximations of extreme tail behaviour, we conduct an Anderson-Darling (AD) test to validate the goodness-of-fit. The AD test is a goodness-of-fit test, in which it estimates whether a sample distribution matches another distribution that you test against. The AD test is also more sensitive to behaviour in the tails, making it appropriate to test the KDE's. As we modelled our KDE's using data from the lower 5th percentile, which are extreme values, the best distribution to model against is the left skewed Gumbel distribution which is used to model minimum extremes. This distribution, in

theory, matches the behaviour of our dataset.

The general form of the Gumbel distribution is:

$$f(x) = \exp(x - e^x)$$

Through the introduction of a location and scaling parameter, you can then obtain the left skewed Gumbel distribution, which is given by:

$$f(x) = \frac{1}{\beta} e^{\left(\frac{x-\mu}{\beta}\right)} \cdot e^{-e^{\left(\frac{x-\mu}{\beta}\right)}}$$

All three of our KDE's passed at a 5% significance level, indicating that the spread behaviour in joint tail events of our sector pairs are well modelled by Extreme Value theory.

Pair	Critical Value	Pass at 5% sf
Technology and Capital Goods	0.722	True
Basic Industries and Capital Goods	0.723	True
Capital Goods and Consumer Services	0.721	True

Table 1: AD Test Against Left-Skewed Gumbel Distribution

3.5 Minimum Spanning Tree

A minimum spanning tree (MST) is a connected, undirected graph that has positive edge weights. MST's therefore, have no cycles and have all vertices connected by a walk. A walk is defined as an alternating sequence of nodes and edges, in which it starts and ends with a node. This MST is implemented using the SciPy 'minimum_spanning_tree()' function, which implements Kruskal's algorithm. It works using a compressed sparse row matrix to iteratively find the smallest edge weight, and connect it with another node to form smaller trees. The process is iterative until all nodes form one tree, connected by the smallest edge weights.

Our MST used the inverse of the mutual information as edge weights as this ensured that sectors with the highest mutual information would be closer to the central node. Therefore implying that the sector pairs with the highest co-movements would be more central. As a result of this, Capital Goods emerged as a highly connected central node, linked to four other sectors: Technology, Consumer Services, Consumer Durables and Basic Industries. This is consistent with our KDE analysis, as we found that Capital Goods was frequently featured amongst the sector pairs ranked by the highest dependency gaps. Three of these four connected nodes also appeared in our KDE analysis. It can thus be said that Capital Goods, being a central node, holds a high degree of influence by acting as a bridge between different clusters. In combination with its consistent appearance in our KDE joint tail event analysis, it can be hypothesised that it is a sector that absorbs risk across the entire network given its constant

underperformance in sector pairs, especially with those centrally connected to it. This raises questions as to possible strategies in dynamic hedging based on exposure to the Capital Goods sector in times of systemic stress.

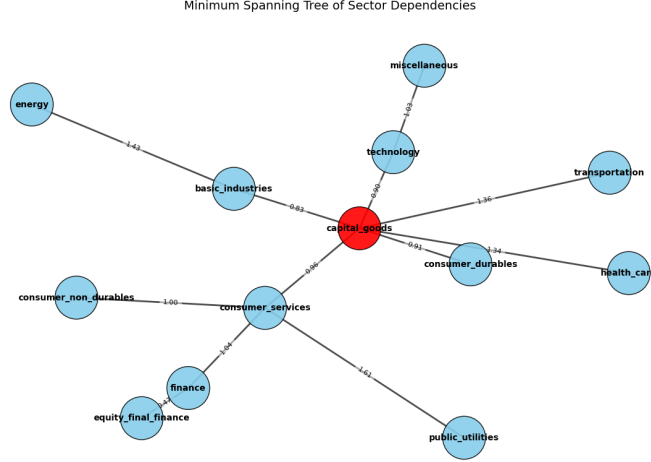


Figure 9: Minimum Spanning Tree

4 Discussion

While this paper touches on the strengths of implementing mutual information as a key tool for tail risk analysis, there are various pathways to build upon this concept. Firstly, while we implemented a simple static MST of our sectors, this could be upgraded to be dynamic as to represent the changing network over time. This would show how sectors can have evolving relationships in different market regimes amongst each other, and would have greater, more practical application in dynamic portfolio hedging.

Another possible expansion on this research would be to implement transfer entropy as to understand the directional flow of information. While we investigated how co-movement, we are not aware of sectors that may be leading or lagging with other sectors. Knowing this allows for further expansion into using sectors as a predictor feature for tail events. This feature could be implemented in machine learning models such as an LSTM to predict future tail risk for use in risk management.

5 Conclusion

To conclude, this project set to explore the complex, non-linear relationships between sector pairs, that cannot be captured by Pearson correlation. By using mutual information, we aimed to extract the non-linear relationship between

two sector returns, and how they evolve in periods of market stress - known as joint tail events.

To focus on co-movement on joint tail events, we filtered returns for days where sectors experienced returns in the lowest 5th percentile. We found that mutual information was able to capture more of the relationship for returns in the 5% quantile than correlation. We then selected three sector pairs that exhibit large dependency gaps - defined as the difference between mutual information and correlation, to model using KDE analysis. KDE allowed us to estimate distributions that were inherently asymmetric, thus revealing non-linear dependencies between sectors in joint tail events.

The results indicate that, mutual information is better at capturing these asymmetric, non-linear dependencies that correlation would otherwise miss. To note, Capital Goods shows that it acts as a possible absorber of risk. It's appearance as a constant under-performer in sector pairs, and its centrality in the minimum spanning tree support the idea. These insights would not be explored if we had solely relied on correlation, as compared to a non-linear dependency measure.

Overall, the findings emphasise the need to incorporate non-linear dependency measures into risk modelling, portfolio optimisation and scenario analysis as well as other fields. A combination of various measures can greatly benefit the understanding of tail risk exposure, especially in areas where linear correlation may not be enough to gather the complexity of relationships between features at hand.

6 Bibliography

References

- Darbellay, G. A. and Wuertz, D. (2000). The entropy as a tool for analysing statistical dependences in financial time series. *Physica A: Statistical Mechanics and its Applications*, 287(3):429–439.
- Dionisio, A., Menezes, R., and Mendes, D. A. (2004). Mutual information: a measure of dependency for nonlinear time series. *Physica A: Statistical Mechanics and its Applications*, 344(1):326–329.
- Embrechts, P., McNeil, A. J., and Straumann, D. (2002). *Correlation and Dependence in Risk Management: Properties and Pitfalls*, page 176–223. Cambridge University Press.
- Guo, X., Zhang, H., and Tian, T. (2018). Development of stock correlation networks using mutual information and financial big data. *PLOS ONE*, 13(4):1–16.
- Maasoumi, E. and Racine, J. (2002). Entropy and predictability of stock market returns. *Journal of Econometrics*, 107(1):291–312. Information and Entropy Econometrics.

Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423.